

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal Sentinel-1 SLC IW mode data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimeter-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 895/2
Bridge Chainage – 894+285

Project - Chittorgarh to Kota
Zone – RJ 06
Structure Type – MNB

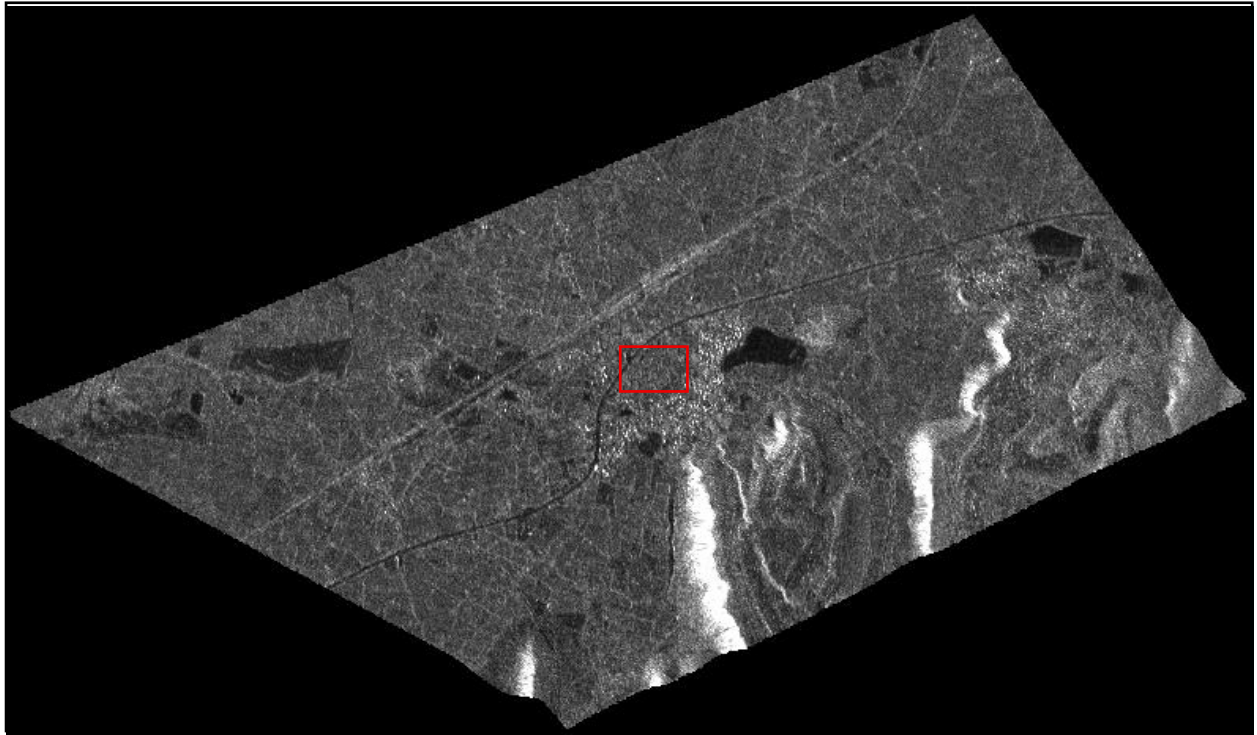
Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

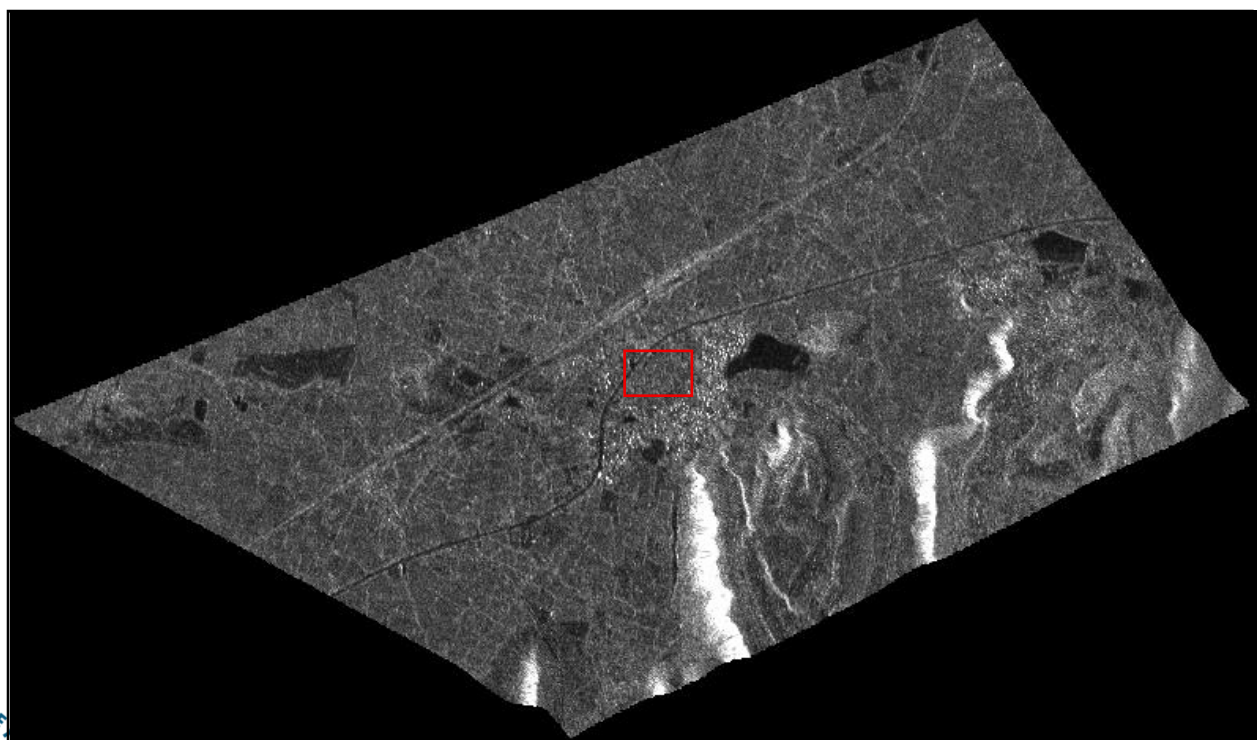
Flight Direction - DESCENDING
Polarization – VV+VH
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization



3. Results of SAR based bridge inspection, including visualizations

Bridge Components

MNB – CH-894+285

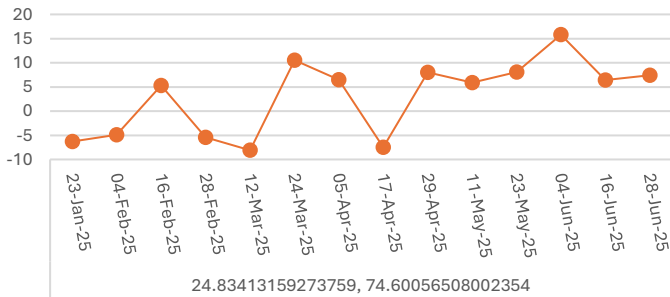
Length – 55.8M



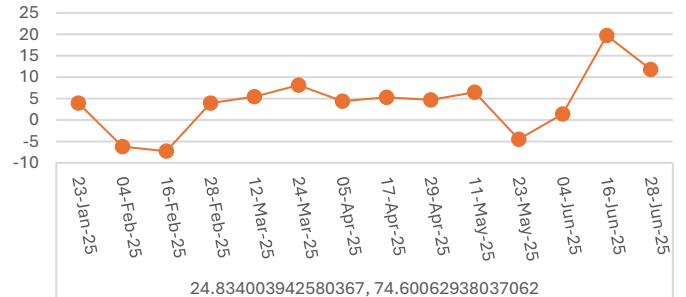
Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
P1		Pier 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

3.1 VV Displacement From 23 Jan 2025 to 28 June 2025

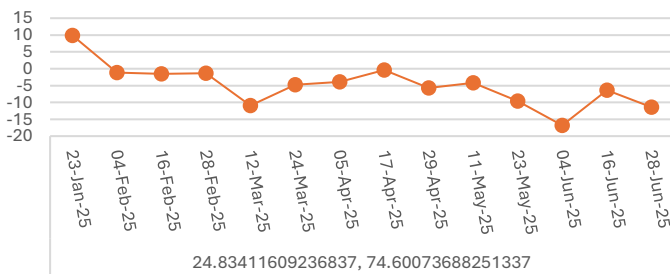
Displacement @A1 LHS VV



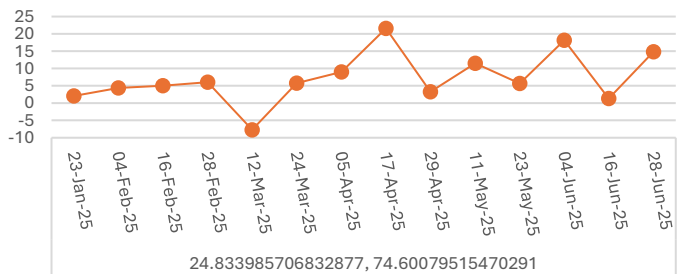
Displacement @A1 RHS VV



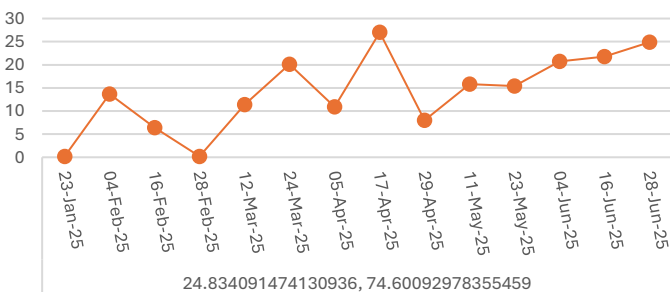
Displacement @P1 LHS VV



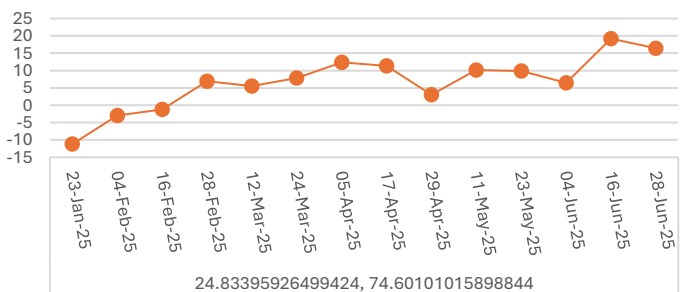
Displacement @P1 RHS VV



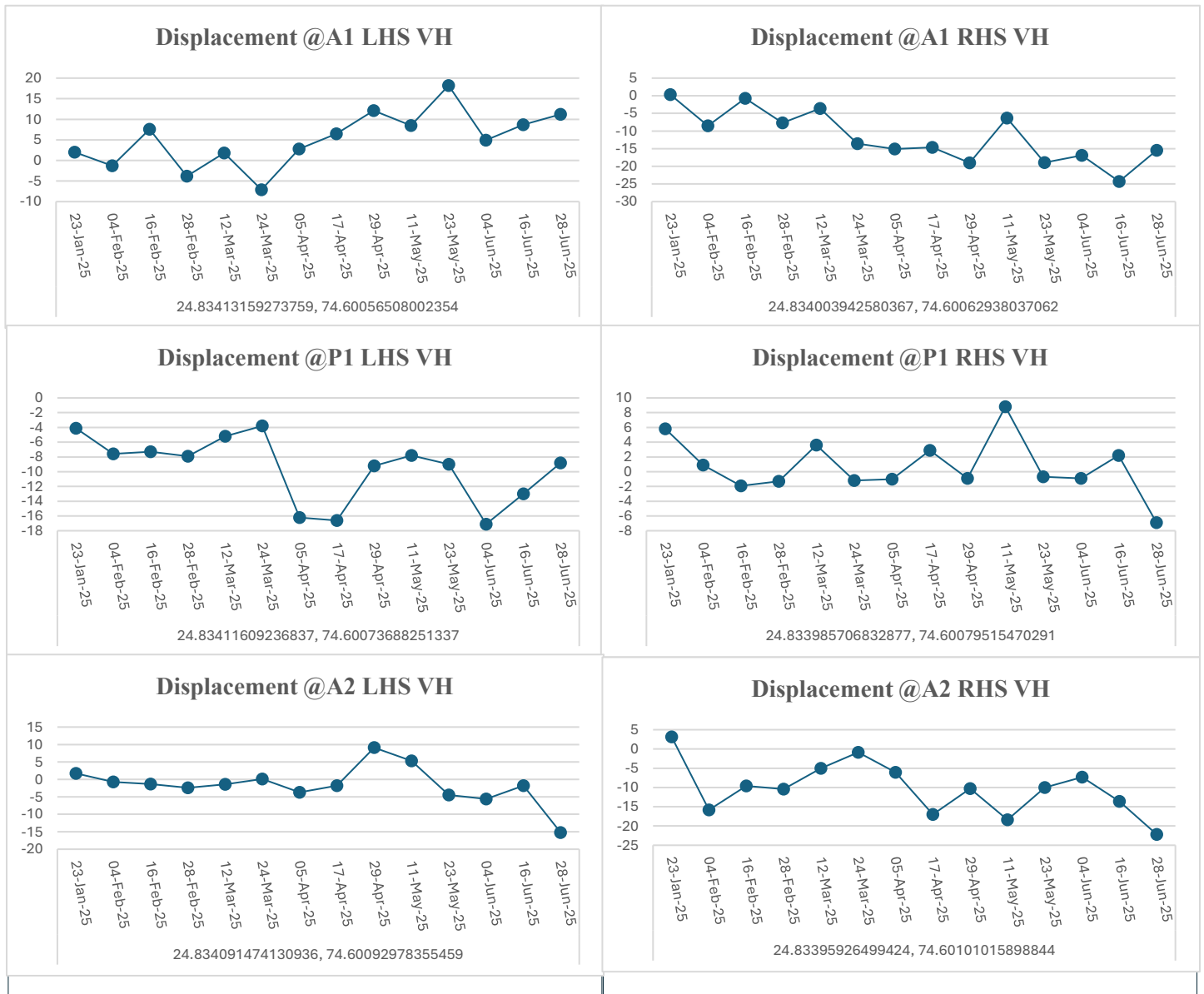
Displacement @A2 LHS VV



Displacement @A2 RHS VV



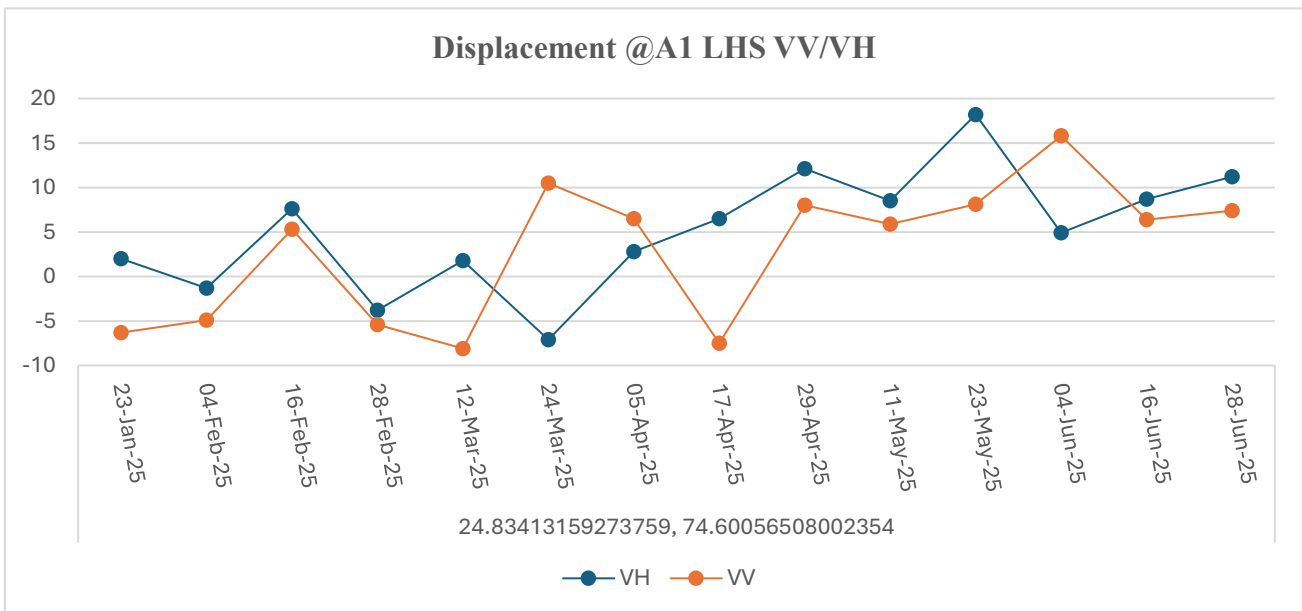
3.2 VH Displacement From 23 Jan 2025 to 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.83413159273759, 74.60056508002354)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** ranges roughly between -8 mm and $+15$ mm, showing alternating uplift and settlement phases.
- **Horizontal displacement (VH)** varies approximately between -4 mm and $+18$ mm, indicating moderate lateral movement across the monitoring period.

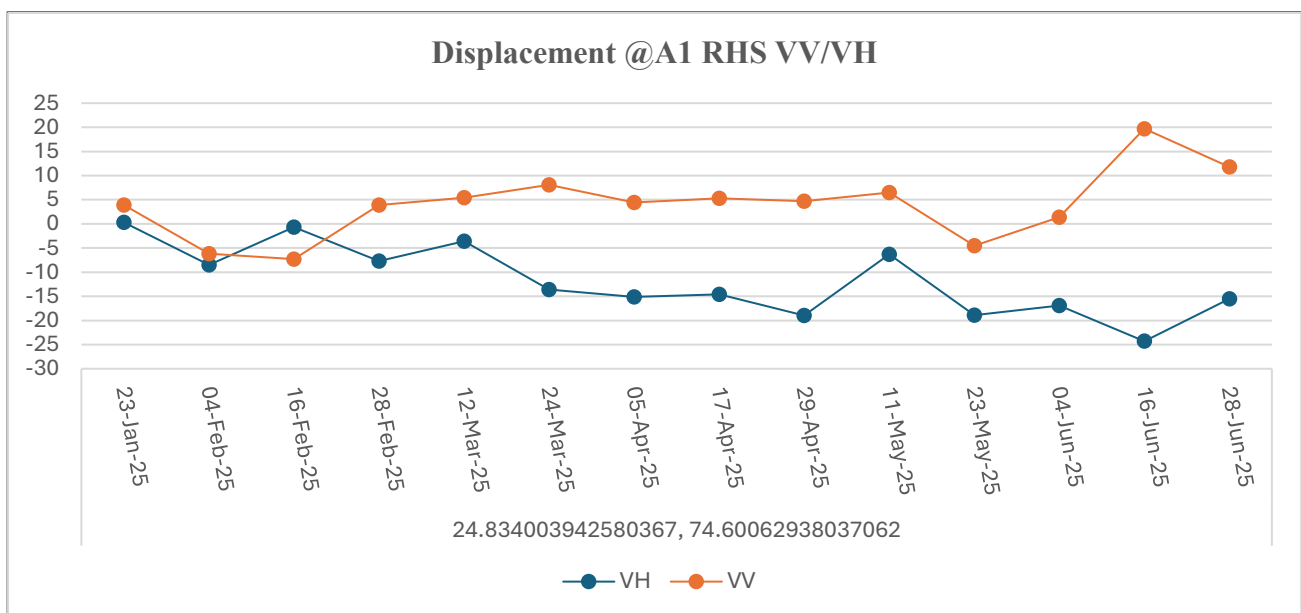
Critical Events & Anomalies:

- **March 2025:** Simultaneous reduction in both VH and VV indicates a short-term disturbance or transient loading condition.
- **May-June 2025:** Peak VH ($+18$ mm) and VV ($+15$ mm) occur close together, suggesting a combined structural response, possibly influenced by thermal expansion or loading variation.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.834003942580367, 74.60062938037062)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement ranges approximately from -25 mm to -5 mm, indicating predominantly negative lateral movement throughout the monitoring period.
- **Vertical displacement (VV)** Vertical displacement varies approximately between -8 mm and +20 mm, showing alternating uplift and settlement.

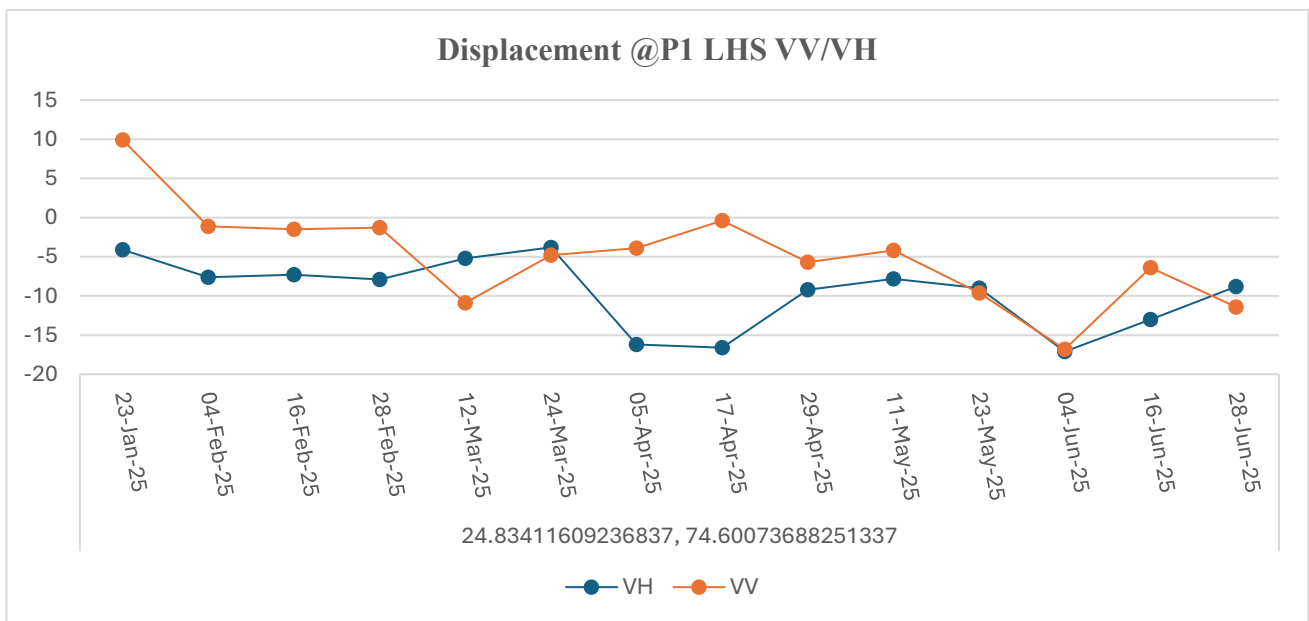
Critical Events & Anomalies:

- **March-April 2025:** Gradual increase in negative VH values indicates progressive lateral movement, possibly related to cumulative loading or foundation response.
- **June 2025:** VV reaches its maximum (+20 mm) while VH remains strongly negative, suggesting combined vertical uplift with lateral drift, which may reflect differential structural behaviour or thermal effects.

3.5 SAR Based Analysis at Pier 1 LHS:



P1 LHS (24.83411609236837, 74.60073688251337)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement remains consistently negative, ranging approximately from -6 mm to -17 mm, indicating sustained lateral movement in one dominant direction.
- **Vertical displacement (VV)** Vertical displacement ranges approximately between -16 mm and +8 mm, showing alternating uplift and settlement phases. Toward the end of the monitoring period, VV partially recovers but remains predominantly negative

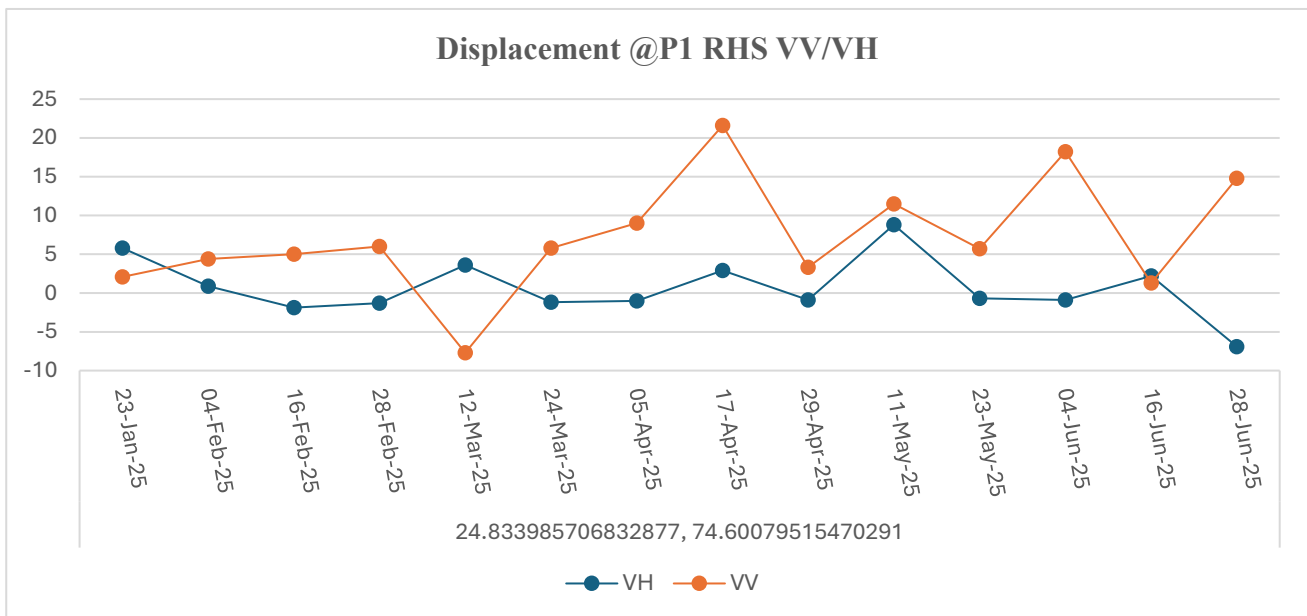
Critical Events & Anomalies:

- **February to March 2025:** A clear downward trend in both VH and VV suggests a period of increased settlement
- **April 2025:** Minimum VH values coincide with continued negative VV, indicating combined lateral and vertical movement
- **June 2025:** Partial recovery in both components suggests stabilization, though full rebound is not observed.

3.6 SAR Based Analysis at Pier 1 RHS:



P1 RHS (24.833985706832877, 74.60079515470291)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement varies approximately between –6 mm and +10 mm, indicating moderate lateral movement throughout the monitoring period.
- **Vertical displacement (VV)** Vertical displacement ranges approximately between –8 mm and +22 mm, showing greater variability compared to VH.

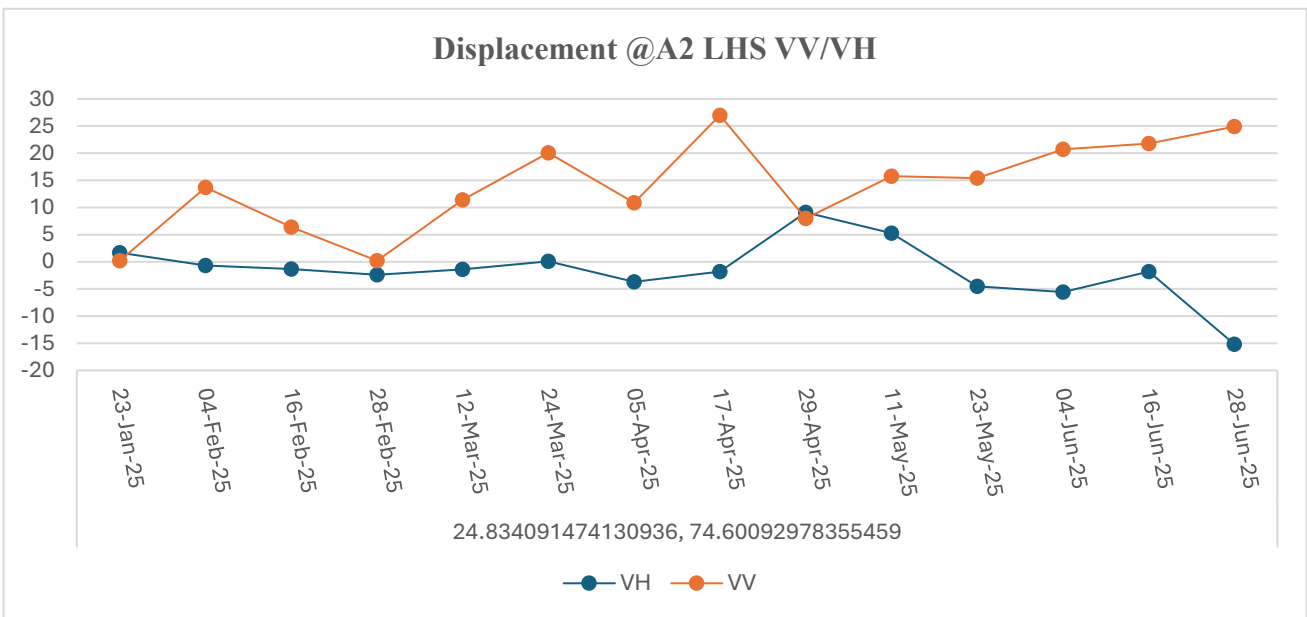
Critical Events & Anomalies:

- **March 2025:** A brief negative VV excursion (–8 mm) indicates short-term settlement prior to upward recovery.
- **April:** Simultaneous peak in VV (~+22 mm) and moderate VH response suggests a combined vertical–horizontal structural response, possibly influenced by thermal or loading effects
- **June 2025:** A second VV peak (~+18 mm) occurs, followed by stabilization, indicating transient rather than progressive behaviour.

3.7 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.834091474130936, 74.6009297835545)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement ranges approximately from -18 mm to +6 mm, indicating a dominantly negative lateral movement over the monitoring period.
- **Vertical displacement (VV)** Vertical displacement varies between approximately +3 mm and +26 mm, showing consistent uplift behaviour.

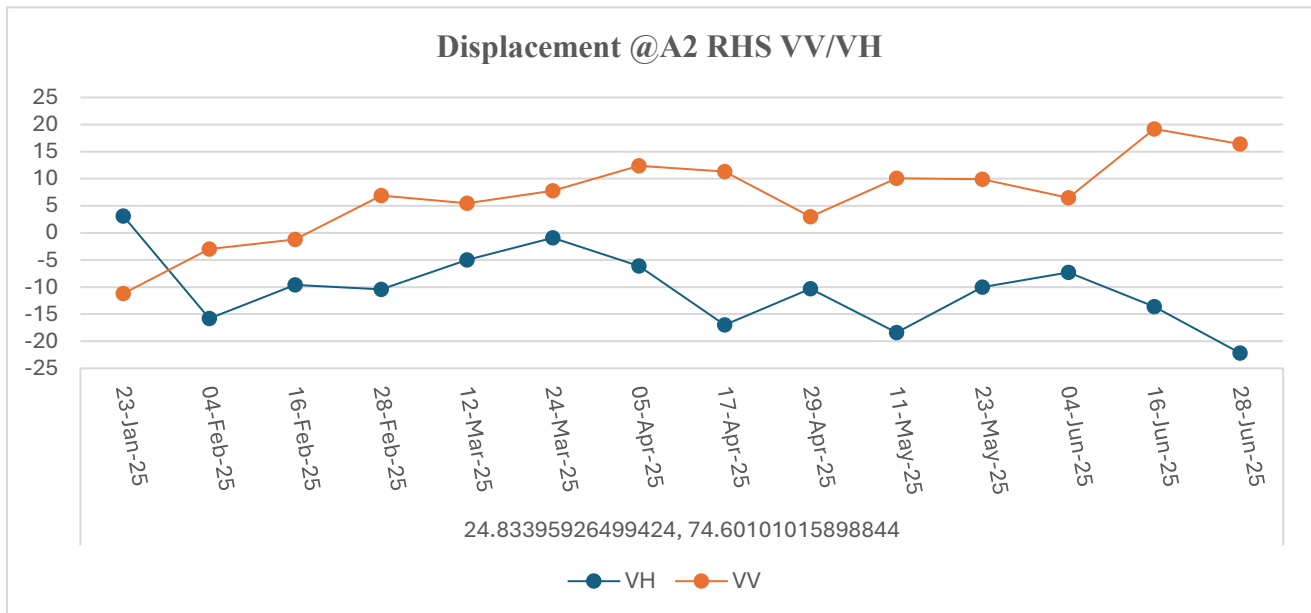
Critical Events & Anomalies:

- **April 2025:** A sharp vertical peak (+26 mm) coincides with a moderate VH recovery, indicating a combined vertical-horizonal structural response.
- **May-June 2025:** VH reaches its most negative values while VV remains elevated, suggesting differential movement potentially influenced by foundation response or thermal effects.

3.8 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.83395926499424, 74.60101015898844)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement ranges approximately from –22 mm to +4 mm, indicating a predominantly negative lateral movement throughout the monitoring period.
- **Vertical displacement (VV)** Vertical displacement varies approximately between –12 mm and +20 mm, showing moderate-to-high variability.

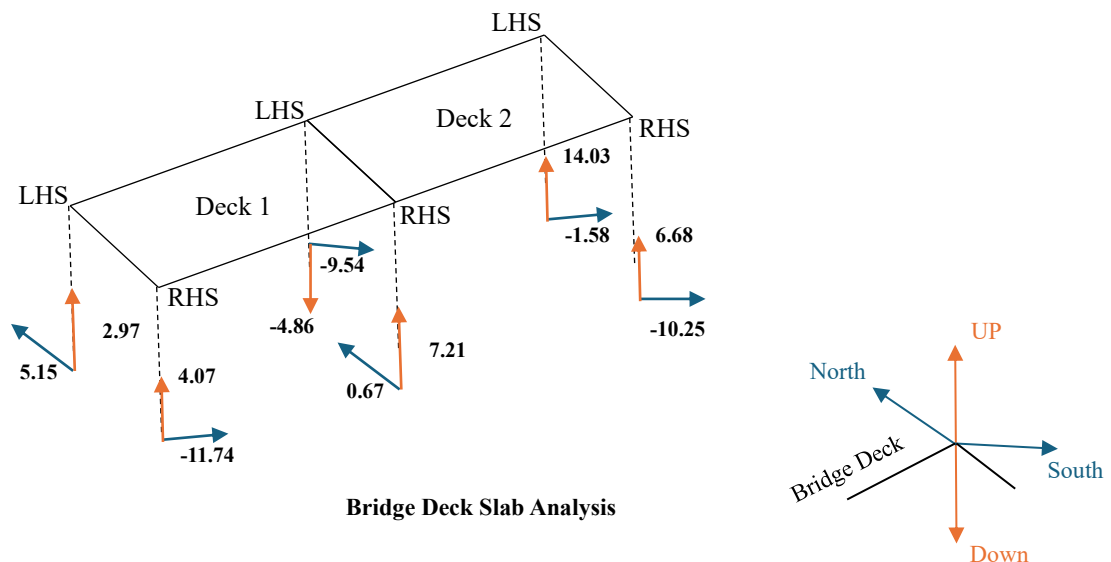
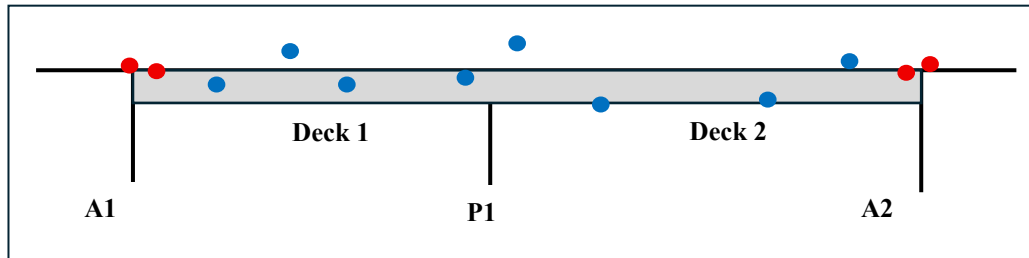
Critical Events & Anomalies:

- **April 2025:** A pronounced vertical uplift (+20 mm) coincides with a notable negative VH excursion, indicating combined vertical uplift and lateral movement.
- **June 2025:** VH reaches its most negative values (–22 mm), while VV remains elevated, suggesting differential structural response and possible redistribution of internal forces.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The bridge deck slab exhibits non-uniform vertical displacement across Deck 1 and Deck 2, indicating differential structural response between spans and support zones. Localised downward displacement is observed at the RHS of Deck 1 (−4.9 mm) and the central lower region (−7.2 mm), indicating partial settlement behaviour within the span.

Horizontal Displacement Behaviour:

- Horizontal displacement demonstrates clear directional variability across the deck system, highlighting asymmetric deformation behaviour. The central region between Deck 1 and Deck 2 shows limited lateral movement (0.7 mm), suggesting a relative pivot or stiffness concentration.

CONCLUSION:

- The bridge deck slab demonstrates a complex, non-symmetric deformation pattern, driven by combined vertical uplift, localized settlement, and horizontal displacement. Deck 2 experiences the highest vertical uplift, while Deck 1 exhibits greater lateral deformation. Overall, the deformation characteristics indicate a structurally active system, warranting continued monitoring to assess progression, particularly at RHS locations and inter-deck interfaces